La Scuola d'Italia / Dr. Huson / IB Math 12 HL
2 November 2026

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5.4 Classwork: Introduction to Differential Equations (no calculator)

Answer Problems 1 and 2 in the spaces provided. Answer Problems 3–5 on lined paper. Show all working and give exact answers.

1. [Maximum mark: 9]

   Consider the differential equation

   $$\frac{\mathrm{d}y}{\mathrm{d}x} = 6 - 2y.$$

   (a) Show that $y = Ae^{-2x} + 3$ is a solution of the differential equation for every constant A. [3]

   (b) Find the particular solution for which $y = 5$ when $x = 0$. [2]

   (c) Write down the value that y approaches as $x \to \infty$, and show that this value of y makes $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$. [2]

   A cup of tea cools in a room at 20°C. The rate at which its temperature T decreases is proportional to the difference between T and the room temperature.

   (d) Write down a differential equation for T in terms of the time t and a positive constant k. [2]

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2. [Maximum mark: 9]

Solve each differential equation by separating the variables. Give each answer in the form $y = f(x)$.

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 6x^2y$, given that $y = 2$ when $x = 0$. [5]

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = e^{x-y}$, given that $y = \ln 2$ when $x = 0$. [4]

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Answer Problems 3–5 on lined paper.

3. [Maximum mark: 9]

The tea in Problem 1 is 90°C when $t = 0$, where t is the time in minutes. Its temperature satisfies

$$\frac{\mathrm{d}T}{\mathrm{d}t} = -k\,(T - 20)\,, \qquad k > 0.$$

(a) By solving the differential equation, show that $T = 20 + 70e^{-kt}$. [4]

(b) After 10 minutes the temperature of the tea is 55°C. Show that $k = \dfrac{\ln 2}{10}$. [2]

(c) Find the time at which the temperature of the tea is 37.5°C. [3]

4. [Maximum mark: 10]

Consider the differential equation $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x + y$, where $y = 1$ when $x = 0$.

(a) Use Euler's method with a step length of 0.5 to find an approximate value of y when $x = 1$. [3]

(b) By using an integrating factor, solve the differential equation. Give your answer in the form $y = f(x)$. [5]

(c) Show that $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 1 + x + y$. Hence explain why the approximation in part (a) is less than the true value of y when $x = 1$. [2]

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5. [Maximum mark: 8]

Consider the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^2 + y^2}{xy}, \qquad x > 0,\ y > 0.$$

(a) Use the substitution $y = vx$ to show that $x\dfrac{\mathrm{d}v}{\mathrm{d}x} = \dfrac{1}{v}$. [3]

(b) Given that $y = 2e$ when $x = e$, solve the differential equation. Give your answer in the form $y = f(x)$. [5]